col = # of unknown

row = # of eqn.

1) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \end{array}\right]$ => Unique soln (each col has leading 1)

2) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 0 & c \end{array}\right]$ => No soln $(0 \neq c)$

$0 = 0$ ✓

3) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 0 & 0 \end{array}\right]$ => infinitely many soln. (zero row).

take x_3 = any no.

$x_3 \in \mathbb{R}$